Disparity analyses are robust to ancestral state estimation uncertainty

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Abstract

Ancestral state estimation is widely used to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty. How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, and both point estimate and probabilistic approaches for treating uncertainty, under continuous and discrete trait models and varying fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

Morphological diversity, hereafter termed disparity, is widely used for investigating macroevolutionary dynamics (Guillerme et al. 2020a). Disparity can be described as the distribution or spread of taxa within a multidimensional trait space defined by morphological similarity (Erwin 2007) and can be quantified using both continuous (García-Navas et al. 2017) and discrete characters (Foote 1994; Lloyd 2016). Morphological trait datasets are routinely integrated with phylogenies when examining disparity across clades or time. These analyses can be used to investigate selectivity across mass extinctions (Bapst et al. 2012; Cole and Hopkins 2021; Liu et al. 2024), can identify timings of key morphological innovations (Foth et al. 2021). However, such analyses are inevitably limited by the incompleteness of the fossil record; only a biased subset of past diversity is preserved, leaving unsampled ghost lineages and obscuring large portions of a clade's evolutionary history (Flannery Sutherland et al. 2019; Smith et al. 2021). This missing data has been shown to result in both under- and over-estimations of disparity (Mitchell 2015; Smith et al. 2023). For example, high amounts of missing data results in 'centroid-slippage', whereby an incomplete taxon is pulled towards the centre of the morphospace (Flannery Sutherland et al. 2019). Conversely, exceptional preservation, such as Lagerstätten taxa may result in inflated disparity, seen in ichthyosaurs (Flannery Sutherland et al. 2019) and pterosaurs (Butler et al. 2012).

One way in which unsampled lineages can be accounted for is through the use of ancestral state estimation (ASE). Ancestral state estimation is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for tips on a phylogenetic tree (Schluter et al. 1997; Cunningham et al. 1998; Revell 2025). ASE has been extensively used across macroevolutionary studies for inferring the likely character state at each node on a phylogeny for continuous or discrete traits of interest. These estimated states can be combined with observed data to generate a more complete trait space, mitigating the gaps left by an incomplete fossil record and, in theory, improve our ability to recover the true disparity signal (Sidlauskas 2008; Brusatte et al. 2011; Lloyd 2018). Inferring ancestral states also allows disparity across time analyses to be explored with higher resolution (Brusatte et al. 2011; Halliday and Goswami 2016; Guillerme 2018).

1.1 Issues with ASE

Despite its theoretical advantages, integration of ancestral state estimation within the disparity analysis pipeline has often been neglected. Much of the apprehension towards the use of ancestral state estimation has likely stemmed from the notion that estimates are inherently uncertain; root-ward estimates in particular are often associated with large confidence intervals or inconclusive scaled likelihoods (Schluter et al. 1997; Losos 1999; Gascuel and Steel 2014). Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al. 2015; Boyko and O'Meara 2024), most computational implementations of ancestral state estimation return point estimates, since there is little guidance on how to correctly treat this uncertainty downstream when constructing the trait space. These point estimates are typically averages from confidence intervals for continuous characters or collapsed states returned from arbitrary thresholds (Smith and Donoghue 2022; Casali et al. 2023). By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses, all while obscuring potentially accurate signals (Ackerly et al. 2006; Vanderpoorten and Goffinet 2006).

Another key issue with ancestral state estimation is its sensitivity to model misspecification. Likelihood-based approaches to ancestral state estimation require both a tree topology and a specified model of character evolution. Estimation of continuous characters are typically modeled under Brownian motion (BM), which describes a directionless, stochastic process where the trait divergence between related taxa is proportional to their shared evolutionary history and the rate of evolution (Felsenstein 1985; Schluter et al. 1997; O'Meara et al. 2006). However, deviations from BM in the trait data, such as directional trends or stabilising selection, can result in inaccurate estimations. For example, both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. Model-related issues can also arise for discrete traits, which are estimated using an Mk model, requiring a tree topology and specified model of evolution (describing transition rates). Estimation accuracy of discrete states appears prone to model over-parameterisation and sensitive to fast transition rates (Keating 2023). Furthermore discrete ancestral state estimation faces a paradoxical trade-off where a slow rate of evolution allows for an easy estimation of ancestral states, but provides little information for estimating transition rates (Gascuel and Steel 2020).

Ané (2008) demonstrated that ancestral state estimation is statistically inconsistent, meaning

that even if the sample size of extant tips were infinite, estimates are not expected to converge on the true value. The inclusion of fossils can improve estimations, due to the shorter evolutionary path to the root of the tree (Finarelli and Flynn 2006; Finarelli and Goswami 2013; Joy et al. 2016; Goswami and Clavel 2025). Numerous simulation and empirical based studies have shown that ancestral state estimations based on extant-only taxa are extremely biased (Webster and Purvis 2002; Finarelli and Flynn 2006; Albert et al. 2009; Betancur-R et al. 2015; Royer-Carenzi and Didier 2016; Monson et al. 2022; Wisniewski et al. 2022; Keating 2023). Similar biases have been observed in disparity analyses (Mitchell 2015). Despite this, many studies are forced to use extant-only trees for both ancestral state estimation (Jetz and Pyron 2018; Amador and Giannini 2021) and disparity (Harmon et al. 2010; Baker et al. 2015; Smith and Donoghue 2022) due to the lack of reasonable fossil data.

1.2 Previous work

Numerous studies have assessed the raw estimation error of ancestral state estimation approaches under different models of evolution, methods of estimation and fossil sampling level (Elliot and Mooers 2014; Royer-Carenzi and Didier 2016; Reyes et al. 2018; Holland et al. 2020; Keating 2023; Boyko et al. 2025). However, how state estimation accuracy is propagated into recovery of an accurate trait space is less understood, despite the continued use of ancestral state estimation in disparity analyses. Lloyd (2018) demonstrated that carrying out ancestral state estimation post-ordination rather than pre-ordination can distort the distribution of points in a trait space markedly, introducing unwanted phylogenetic signal and low levels of convergence. However, this study was restricted to a single empirical dataset and to discrete traits. Thus, we lack a comprehensive understanding of how estimation approaches, trait data and fossil sampling level interact to affect our ability to recover accurate patterns in trait space.

1.3 Aims

Using a simulation framework, we expand on this work to evaluate the ability of different ancestral state estimation approaches to recover true disparity values across different evolutionary models and fossil sampling regimes. Simulations offer a robust approach for benchmarking methodologies since a ground-truth dataset can be generated, so trait values and the resulting disparity signal are all known *a priori* (Lotterhos et al. 2022). Different methodological choices and estimation scenarios can then be evaluated against the true disparity value. Specifically, we directly compare

pre- versus post-ordination estimation approaches, as well as point-estimate versus probabilistic treatments of uncertainty, thereby examining two axes of methodological variation. We also investigate how estimation error varies under different models of trait evolution and fossil sampling regimes. This integrative framework provides a comprehensive evaluation for understanding when ancestral state estimation improves disparity estimation, when it introduces bias, and how uncertainty should be treated when working with incomplete fossil data.

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

All analyses were carried out using R 4.5.2 (R Core Team 2025). We simulated birth-death trees with the speciation (λ) parameter set to 1.0, and extinction (μ) set to 0.9 (Guillerme 2024). Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. These parameters correspond to a high turnover regime, consistent with the upper-empirical estimates from diversification studies of animal groups (Scholl and Wiens 2016; Pires et al. 2018; Henao Diaz et al. 2019; Morlon et al. 2024). This level of extinction produces realistic evolutionary histories with substantial extant and extinct diversity, while maintaining a positive net diversification rate ($\lambda - \mu > 0$).

We generated three sets of 100 birth-death trees with these parameters until the simulations reached 50, 100 and 150 living taxa, with maximum tree size of 500, 1000 and 1500 tips, respectively. The randomness of the birth-death model meant some simulated trees contained only extinct taxa, so we re-ran each individual simulation until we obtained the desired number of living taxa (50, 100 or 150). This meant that although the parameters were set to $\lambda = 1$ and $\mu = 0.9$, the effective parameters slightly deviated from these input values (see Supplementary Figure 1). Mean net diversification across all simulated trees was 0.30 ± 0.09.

Trees were pruned to extract the crown clade of the tree for downstream analyses. This ensured that all simulated fossil sampling levels shared an identical root node and tree height, providing a consistent framework for downstream analyses. Any tip included in the analyses was a descendant of the most basal node considered, i.e., the MRCA of extant tips, guaranteeing that all ordinations and disparity values were calculated within a consistent clade, allowing a robust assessment of the interaction between fossil sampling and ancestral state estimation.

2.1.2 Trait simulations

For each phylogeny, we simulated continuous and discrete trait data separately. Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (table 1). We simulated BM with variance = 0.25. We also simulated BM with a trend (BM_t), where the ancestral trait value increased by an increment of 0.1 multiplied by branch length. For OU models, the data were simulated with strong attraction to the optimum (α) (OU_{st}), weak α (OU_w) and with differing optimum and root states (OU_{sh}). Following Gearty et al. (2024), the α parameter was set based on phylogenetic half life ($\frac{ln(2)}{\alpha}$) to account for differing tree heights across simulations (Cooper et al. 2016). For the strong and weak OU models, α was set as 25% and 75% of the tree height respectively. We simulated 20 independent characters for each model (BM, BM_t, OU_w, OU_{st}, OU_{sh}) across each phylogeny.

Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to the product of the transition probability and branch length. We simulated discrete traits across three rate categories, representing fast (0.9), medium (0.1) and slow (0.01) transition rates (table 1). These rates were assigned by generating equal-rates (ER) transition probability matrices. For each category, 85 binary characters and 15 three-state characters were simulated, following a typical scoring ratio found in empirical data (Guillerme and Cooper 2016; Casali et al. 2023).

2.2 Fossil sampling simulation

To simulate different levels of fossil sampling, extinct tips were either retained ('fossilized') or pruned using a Bernoulli sampling method at five different probabilities: 0%, 5%, 15%, 50% and 100%. This method means each extinct tip had an equal probability of being 'fossilized' at each sample level, irrespective of the total tree size. The corresponding trait matrices were subset to match each downsampled tree.

2.3 Ancestral state estimation (ASE)

We estimated ancestral state on the down-sampled matrices using the `dispRity::multi.ace` (Guillerme 2018), which is a wrapper for `ape::ace` (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering the true disparity signal. These were then compared to a null model bypassing ancestral state estimation (Fig. 2, Step 4).

2.3.1 Pre-ordination estimation

Ancestral state estimation of trait data can be carried out at two points either side of the ordination step in the disparity pipeline (Lloyd 2018). Pre-ordination estimation (pre-OASE) involves the estimation of the raw trait values for each node. Post-ordination estimation (post-OASE) calculates the continuous scores across each PC axis of an ordination matrix.

Pre-OASE of continuous characters models evolution under a Brownian motion model, where ancestral state values lie within a multivariate normal distribution that is estimated given the observed tip data and the shared evolutionary history among taxa (Schluter et al. 1997; Rohlf 2001). The width of estimated confidence intervals reflects the uncertainty in continuous ancestral estimates, which typically increases with node depth and decreases with the number of informative descendant lineages.

The pre-OASE method for estimating ancestral discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a maximum likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

We derived trait spaces from which to estimate disparity through ordination of these estimations. For discrete characters, distance matrices were computed using the Maximum Observed Rescaled Distance metric (MORD) (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA) (`cmdscale()` function in R). Continuous data were ordinated using principal components analysis (PCA) (`prcomp()` function in R).

2.3.2 Post-ordination estimation

For post-OASE, the same ordination step was applied prior to ancestral state estimation to the down-sampled discrete and continuous character matrices to generate a set of tip-only trait spaces. Ancestral node states were then estimated under a Brownian motion model across all PC axes, as detailed in section 2.3.1.

2.3.3 Uncertainty treatment

For each estimation method, we also compared point estimate and probabilistic approaches. The point estimate approach returns a fixed single value for each estimate. For continuous

characters, the point estimate returned was the median of the ancestral estimate confidence interval. For discrete characters, we used a strict majority rule where the state with the highest scaled likelihood was selected at each node for each character. In scenarios where state likelihoods were equal, one state was chosen at random. We also used a relative majority rule, where any state at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states was collapsed, otherwise states were left as uncertain (supplementary).

The probabilistic approach differs in that uncertainty in estimates are explicitly accounted for by sampling multiple results rather than a single point estimate. For continuous characters, we sampled each state across 100 matrices, with each estimate value lying within a normal distribution centered on the mean ancestral estimation and spread across the estimated confidence interval. Each discrete character state at each node was sampled 100 times according to their scaled likelihoods (resulting in 100 character estimation matrices). In a scenario where there is uncertainty in a binary character for a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this distribution approach results in a 55% probability of sampling state 0 in each matrix for the respective node (and 45% chance of sampling state 1). Both procedures were implemented using the `sample = 100` argument in the `dispRity::multi.ace` function (Guillerme 2018).

2.4 Disparity analyses

To ensure fair comparison among different fossil sampling levels, only the nodes present across all fossil sampling levels were retained from the ancestral state estimation outputs for each phylogeny; therefore, comparisons in disparity calculations among fossil sampling levels differed solely on the difference in number of fossil tips and the estimations of node states.

When assessing overall disparity, we wanted to understand how well ancestral state estimation can recover true patterns in size and density in trait space occupancy (Guillerme et al. 2020b). Size-based metrics describe the spread of points in the trait space (Guillerme et al. 2020b). To explore how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles calculates the range on each PC axis for the 95% of the data and then sums these ranges across all axes. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992), but with less sensitivity to outliers and sample size. For density, we used the sum of variances (Ciampaglio et al. 2001; Wills 2001) and the average nearest neighbour distance (Foote 1992). Disparity values were calculated across each ancestral state estimation and the true simulated values so that a relative disparity

error could be calculated (Fig. 2 Step 5).

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \tag{1}$$

We calculated disparity across three different treatments of the ordinated matrices. First, we used the untreated ordination output to examine how well ancestral state estimation recovers the true disparity signal across all PC axes. Second, we min-max standardised the PC axes so that the absolute magnitude in PC scores did not influence the disparity metric scores (i.e. in this case, the max-score for PC1 was always standardised to 1). Third, we only used the minimum number of dimensions that were contained across all fossil sampling levels for that specific phylogeny so that each fossil sampling level, i.e., the first 48, 98 and 148 dimensions for the 50, 100 and 150 living taxa simulations, respectively. This was to negate any effect that higher-dimensional trait space estimations were more likely to be closer the true disparity signal.

2.5 Statistical tests

Relative disparity error scores were made absolute and log transformed, so that the lower the value, the closer to 0 the original error (and therefore more accurate). We used ANOVA to explore the effect and interaction that the estimation method, transition rate/model and fossil sampling level had on disparity error. Given that the distribution estimation methods (pre-OASE and post-OASE) had a much larger sample size than the point estimate methods, we used the median of the distributions. Then Tukey test.

3 Results

3.1 Effect of estimation method

For discrete characters, the pre-ordination estimation (pre-OASE) methods were more accurate at recovering the true disparity value than post-ordination estimation (post-OASE) methods. In particular, the pre-OASE distribution method was the best performing ancestral state estimation method. The null model of bypassing ASE consistently recovers an underestimate of disparity.

3.2 Effect of model of evolution

For discrete characters, disparity outputs from slower rates of character evolution were easier to recover than faster rates of evolution. In general, each estimation method performed better at recovering slow rates than fast rates,

3.2.1 Discrete

3.3 Overall Performance of ASE Methods

4 Discussion

5 Conclusions

Acknowledgments

Author Contributions

Data Availability

Competing Interests

Figures

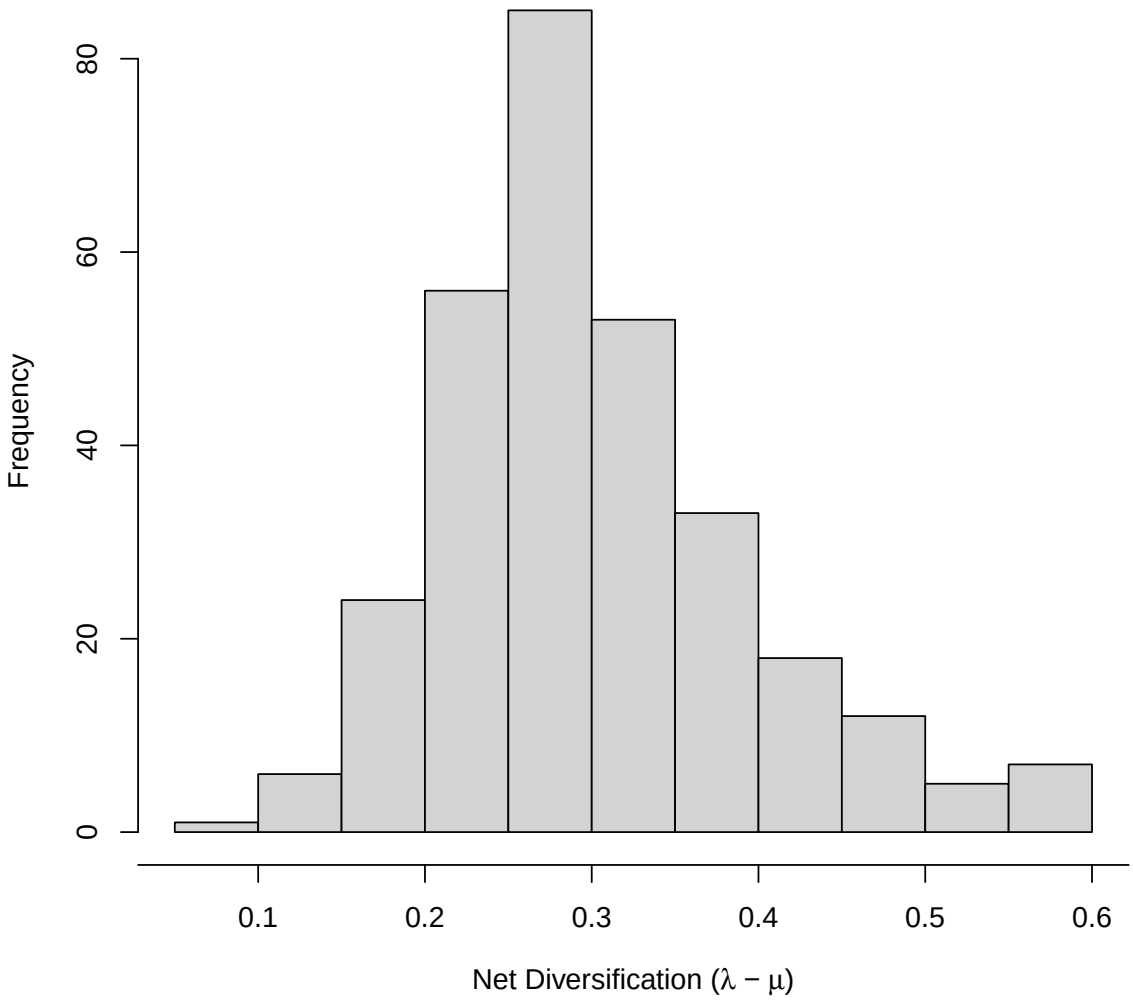

Figure 1: **Birth-death parameters for tree simulation (overall disparity)** Distribution of net diversification values for all trees simulated.

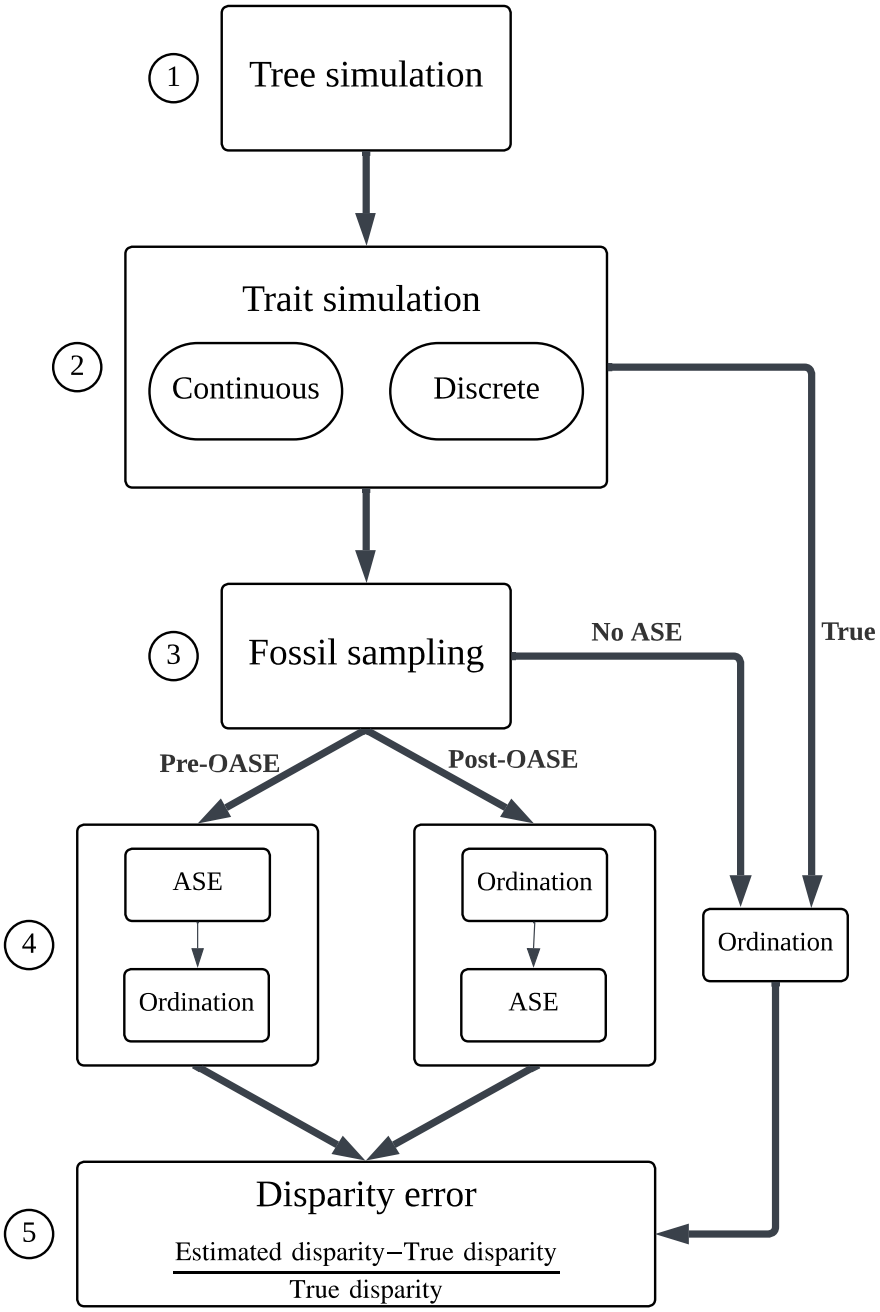

Figure 2: **Simulation workflow**

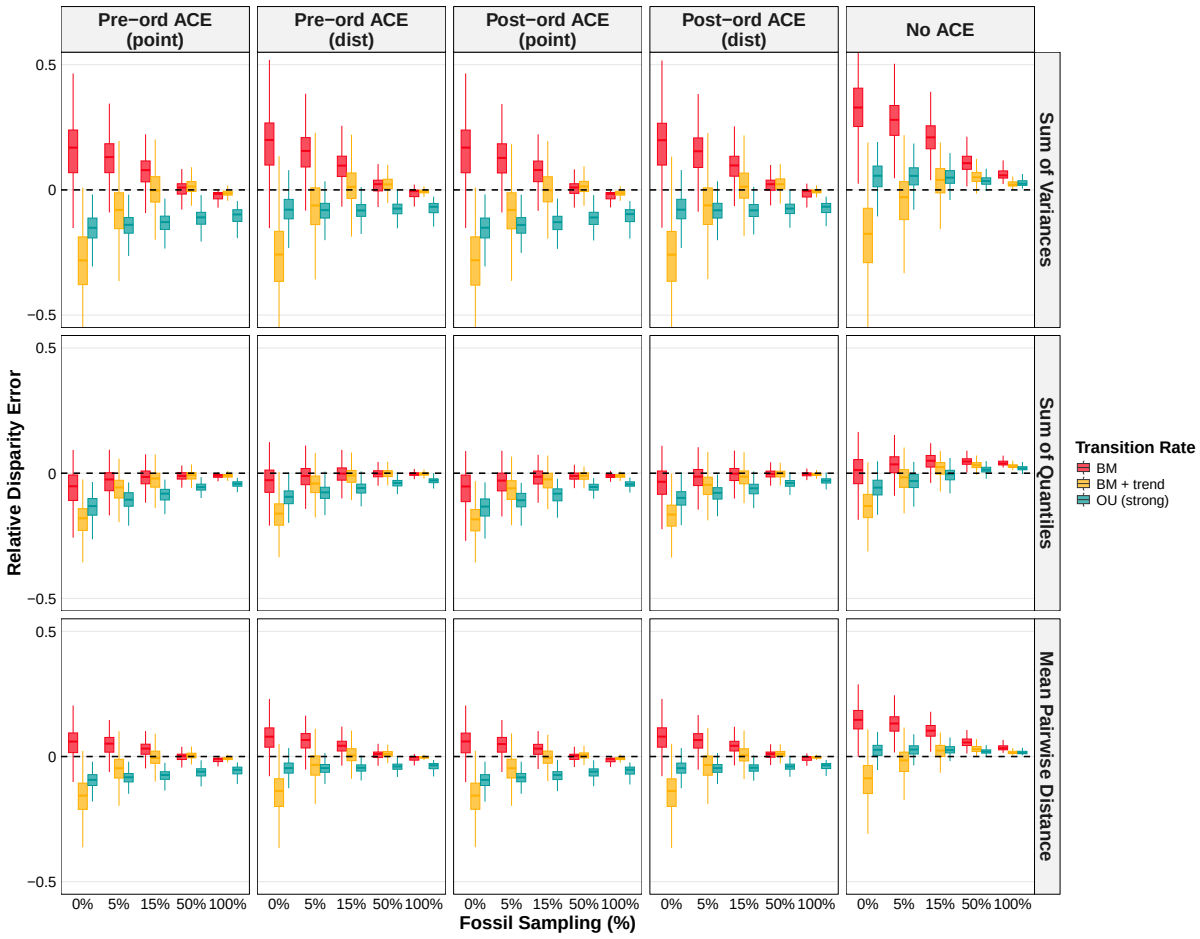

Figure 3: **Results continuous** Continuous

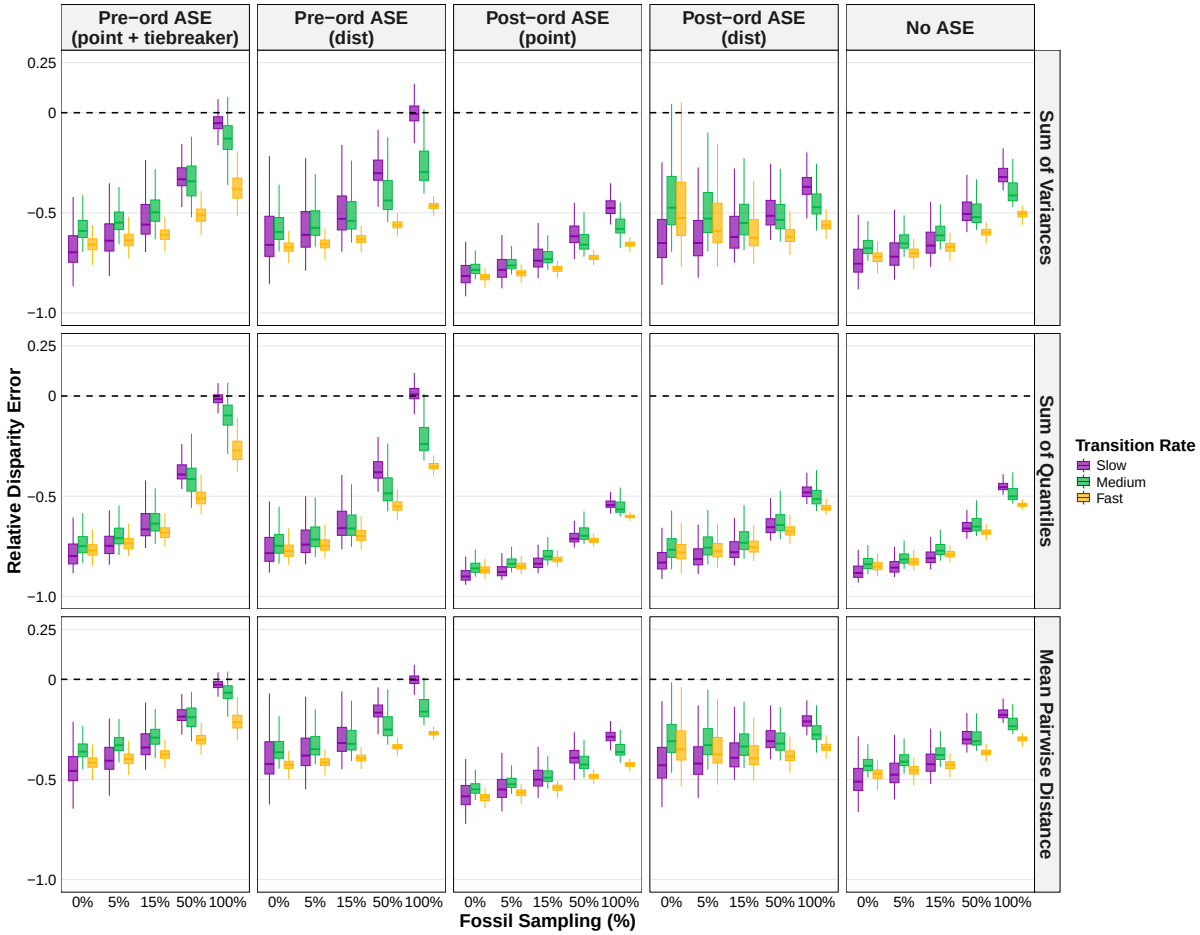

Figure 4: **Results discrete (raw)** Relative disparity errors across all dimensions returned by ordinations.

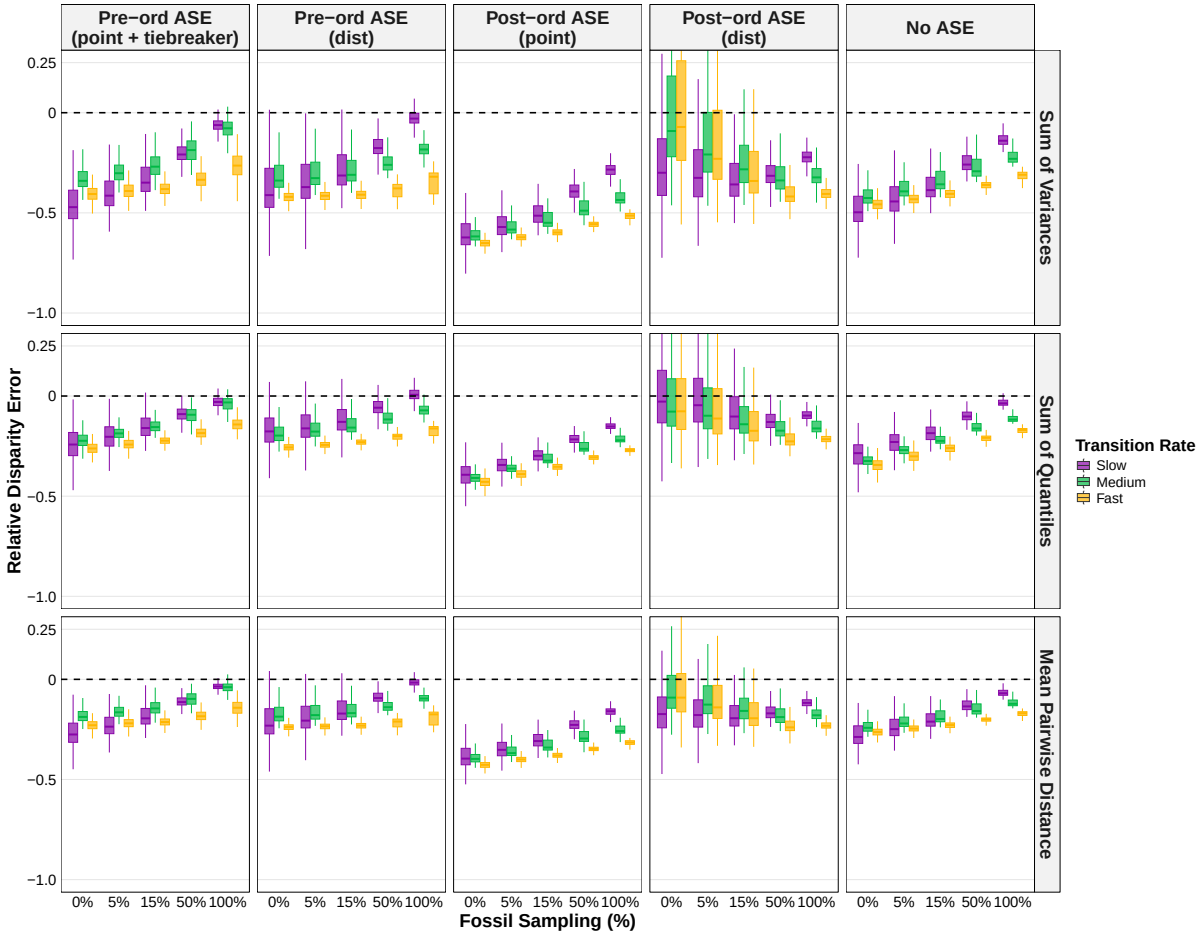

Figure 5: **Results discrete (equal axes)** Relative disparity errors across minimum number of shared dimensions across fossil sample levels for each tree size.

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0. P_{ij} = transition probability; σ = variance; α = rate of adaptation toward the optimum; θ = optimum.

Type of data	**Model**	**Parameters**
Discrete	$Fast$	$P_{ij} = 0.9$
	$Medium$	$P_{ij} = 0.1$
	$Slow$	$P_{ij} = 0.01$
Continuous	BM	$\sigma = 0.25$
	BM_t	$\sigma = 0.25,\ \text{trend} = 0.1$
	OU_{st}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.25}$
	OU_w	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75}$
	OU_{sh}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75},\ \theta = 2$

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